MAT2040
Tutorial 6

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Quick Review

1. Vector Space (I)
   - generalized operations
   - axioms
   - polynomial, function
2. Vector Space (II)
   - properties
   - subspace
3. Basis & Dimension
   - basis
   - dimension
   - coordinates

Question 1

Show that the nonzero rows of a row-echelon matrix R are independent.

Solution 1

We illustrate the case with 3 leading 1s; the general case is analogous. Suppose R has the form

$$R = \begin{bmatrix} 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where $*$ indicates a nonspecified number. Let R_1, R_2, and R_3 denote the nonzero rows of R. If $t_1R_1 + t_2R_2 + t_3R_3 = 0$, we show that $t_1 = 0$, then $t_2 = 0$, and finally $t_3 = 0$. The condition $t_1R_1 + t_2R_2 + t_3R_3 = 0$ becomes

$$(0, t_1, *, *, *, *)+(0, 0, 0, t_2, *, *)+(0, 0, 0, 0, t_3, *) = (0, 0, 0, 0, 0, 0)$$

Equating second entries show that $t_1 = 0$, so the condition becomes $t_2R_2 + t_3R_3 = 0$. Now the same argument shows that $t_2 = 0$. Finally, this gives $t_3R_3 = 0$ and we obtain $t_3 = 0$.

Question 2

Let $\mathbb{R}[x]$ be the vector space of all real polynomials in x. Determine whether the following sets are subspaces of $\mathbb{R}[x]$. Justify your answer.

(a) All polynomials $f(x)$ of degree ≥ 3.

(b) All polynomials $f(x)$ satisfying $f(1) + 2f(2) = 1$.

(c) All polynomials satisfying $f(x) = f(1 - x)$.

Solution 2

(a) Let $W = \{f(x) \in \mathbb{R}[x] : \text{degree } f(x) \geq 3\}$. It is not a subspace since $x^3 + x, -x^3 \in W$ but $\text{degree}((x^3 + x) - x^3) = \text{degree } x = 1 < 3$ and hence $(x^3 + x) + (-x^3) \notin W$.

(b) Let $W = \{f(x) \in \mathbb{R}[x] : f(1) + 2f(2) = 1\}$. Since $f(1) + 2f(2) = 0 \neq 1$ for $f(x) = 0$, $0 \notin W$. Therefore, W is not a subspace.

(c) Let $W = \{f(x) \in \mathbb{R}[x] : f(x) = f(1 - x)\}$. Since $f(x) = f(1 - x) = 0$ for $f(x) = 0$, $0 \in W$. For $f_1(x), f_2(x) \in W$ and $c \in \mathbb{R}$,

$$\begin{cases} f_1(x) = f_1(1 - x) \\ f_2(x) = f_2(1 - x) \end{cases} \rightarrow f_1(x) + cf_2(x) = f_1(1-x) + cf_2(1-x)$$

Therefore, $f_1(x) + cf_2(x) \in W$ for all $f_1(x), f_2(x) \in W$ and all $c \in \mathbb{R}$. Consequently, W is a subspace.

Question 3

Please determine the dimensions of the spaces defined as follows:

(a)

$$\text{Span}\left\{\begin{bmatrix}1\\-2\\2\end{bmatrix}, \begin{bmatrix}2\\-2\\4\end{bmatrix}, \begin{bmatrix}-3\\3\\6\end{bmatrix}\right\}$$

(b)

$$\text{Span}\{(x-2)(x+2), x^2(x^4-2), x^6-8\}$$

Solution 3

(a) Verify if three vectors are linearly independent:

$$\begin{bmatrix} 1 & 2 & -3 \\ -2 & -2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -3 \\ 0 & 2 & -3 \\ 0 & 0 & 12 \end{bmatrix}$$

So dim $= 3$.
(b) Check if one of the three polynomials can be represented by the rest. See if such $k_1, k_2 \in \mathbb{R}$ exist:

$$x^6 - 8 = k_1(x-2)(x+2) + k_2x^2(x^4-2)$$

$$= k_1(x^2-4) + k_2(x^6-2x^2) = k_2x^6 + (k_1-2k_2)x^2 - 4k_1$$

We can get the solution $k_1 = 2, k_2 = 1$.
So one of the elements can be represented by the other two.
Thus dim $= 2$.

Question 4

Find the basis for

$$V = \text{Span}\left(\begin{bmatrix}1\\0\\-3\\3\\2\end{bmatrix}, \begin{bmatrix}3\\1\\0\\4\\0\end{bmatrix}, \begin{bmatrix}1\\1\\6\\-2\\-4\end{bmatrix}, \begin{bmatrix}3\\0\\-1\\1\\2\end{bmatrix}\right).$$

Solution 4

Using elementary row operations, rewrite A in row-echelon form as follows:

$$B = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We can conclude that the basis for A is

$$w_1 = \begin{bmatrix} 1 \\ 0 \\ -3 \\ 3 \\ 2 \end{bmatrix}, \quad w_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 4 \\ 0 \end{bmatrix}, \quad w_3 = \begin{bmatrix} 3 \\ 0 \\ -1 \\ 1 \\ 2 \end{bmatrix}.$$